Answer Partitions, Oracle Response Spectra, and Exact Quantum Query Advantage

Karl Svozil *
Institute for Theoretical Physics, TU Wien, Wiedner Hauptstraße 8–10/136, 1040 Vienna, Austria
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An oracle problem pairs an answer partition, which says which hidden instances share an output, with an access model, which says what a query returns. We show that the access model changes exact quantum query complexity in ways that classical point values cannot detect. Exact one-query resolution is block discrimination of the oracle family, and we state the criterion together with a dual certificate. For point access through a response unitary V, one query resolves a dictator exactly if and only if 0 lies in the convex hull of the spectrum of V, and a two-bit parity—including Deutsch's problem—if and only if -1 is an eigenvalue of V; no other task is one-query exact. Two reversible completions of the same blank-register table, a qutrit transposition and a cyclic shift, therefore have exact costs one and two, and a phase-dressed shift restores cost one; with bounded error the shift still succeeds with the optimal probability 7/8 after one query. A sign oracle cannot resolve any partition that separates a truth table from its complement, and an inaccessible witness of k address bits raises the quantum cost of Bernstein–Vazirani identification from one query to $\min\{k+1, n\}$ while leaving every classical value intact. Holding the standard point oracle fixed, we reorganize the known exact complexities of the 35 balanced three-bit partitions and classify all 6435 balanced four-bit partitions exactly; rational dual certificates settle the 6040 four-bit cases that previously had only numerical support.

I. INTRODUCTION

Deciding whether a one-bit function is constant or balanced takes one quantum query to the standard XOR oracle, against two classical queries [1–3]. A qutrit oracle that returns exactly the same classical values, the cyclic shift $y \mapsto y+b_x \bmod 3$, needs two quantum queries for an exact answer, whereas a qutrit transposition that agrees with it on a blank register needs only one (Sec. III). What decides the difference is the spectrum of the oracle's response, which no classical query can see.

Such examples are instances of a general fact. A query problem consists of an answer partition, which groups the hidden instances into cells whose members share the required answer, and an access model, which says what each query returns, classically or quantum mechanically; query complexity, and with it any quantum speedup, is a property of this task–oracle pair. That it depends on the access model is known in general: simulating an in-place oracle for a permutation of N points can require $\Theta(\sqrt{N})$ uses of the standard oracle for the same function [4]; the comparison of quantum and classical oracle power depends on how the oracle is represented [5]; a richer classical oracle solves Deutsch's problem with one query [6]; and a faulty oracle removes Grover's speedup [7].

Earlier work, including the author's, formulated quantum decision problems as partitions or identifications of prepared quantum states [8–13]. In the query model the partition lives instead on black-box transformations, and it must first be translated into orthogonal output sectors by the oracle and the protocol. This paper asks which features of the access, beyond the classical values it returns, decide the exact quantum cost of that translation.

The translation proceeds in four steps, used throughout the paper and listed in Sec. VI. First, the task fixes the answer partition: instances with the same correct answer share a cell, and only the cell must be learned. Second, the access model fixes what one oracle call does. Third, an instance-independent probe, typically a superposition fanned out from a single basis state, is sent through the oracle, which turns it into an output state that depends on the hidden instance. Fourth, one call solves the task exactly if and only if some probe makes the output states of different cells orthogonal; overlaps within a cell do no harm. In the example above, this orthogonality needs a relative factor -1 between interfering branches, which the flip and the transposition supply and the cyclic shift does not.

The main new result is a spectral criterion. For point access through a response unitary V, one query resolves a dictator exactly if and only if 0 lies in the convex hull of the spectrum of V, and a two-bit parity—including Deutsch's problem—if and only if -1 is an eigenvalue of V; no other task is one-query exact (Theorem 1 and Proposition 4). For the XOR oracle this reduces to the characterization of Chen, Ye, and Li [14]. Its consequences are that two reversible completions of the same blank-register table have different exact costs, that a phase invisible to basis queries can restore the advantage, and that an odd cyclic response keeps a bounded-error advantage although it loses the exact one (Sec. III). The second new result is an exact classification of all balanced four-bit tasks under the standard point oracle: rational dual certificates prove the two-query lower bounds that were previously supported only numerically [15] (Sec. IV). The remaining sections supply the framework—a block form of standard discrimination conditions, together with its dual (Sec. II)—and a decohered-access example in which an inaccessible address wit-

* karl.svozil@tuwien.ac.at

ness changes Bernstein–Vazirani costs while leaving every classical value intact (Sec. V). Table I separates prior results from the present additions.

These results also sharpen the usual intuition about quantum parallelism: the resource that matters for the separations below is the structure of the oracle response—its phases, spectrum, accessible registers, and environment—rather than the number of branches in a superposition.

II. ANSWER PARTITIONS AND EXACT ONE-QUERY RESOLUTION

Let Ω be a finite instance set and let $g : \Omega \to J$ specify the required answer. The task induces the *answer partition*

$$\Pi_g = \{C_j : j \in g(\Omega)\}, \qquad C_j = g^{-1}(j). \tag{1}$$

This partition is indispensable bookkeeping, but it is not yet an access model or a complexity statement. In the query model the hidden instance labels a black-box transformation; the algorithm is not handed a state $|a\rangle$ on which the answer partition is a pre-existing observable. The operational question is whether the specified oracle and a fixed protocol can transfer the cell label to an accessible output.

Let $\rho_a^{(T)}$ be the state of the registers available to the final measurement, produced from instance a after T queries. Exact resolution means that there is one instance-independent measurement $\{P_j\}$ with

$$P_j P_\ell = \delta_{j\ell} P_j, \quad \sum_j P_j = I, \quad \mathrm{Tr}(P_{g(a)} \rho_a^{(T)}) = 1. \tag{2}$$

Supports belonging to different cells are then orthogonal; no orthogonality is required within a cell. The C_j remain sets of classical instances. They are not subspaces, and the answer sectors $P_j \mathcal{H}_{\mathrm{out}}$ arise only after an oracle and a protocol have encoded them. The partition therefore specifies a required zero pattern in the output distinguishability matrix rather than a ready-made physical observable.

We write $Q_\epsilon(\Pi;\mathcal{U})$ and $R_\epsilon(\Pi;\mathcal{U}_{\mathrm{cl}})$ for the optimal quantum and classical worst-case numbers of calls needed with worst-case error at most ϵ, under explicitly matched access conventions, $Q_E \equiv Q_0$ for exact quantum complexity, and D for the depth of an optimal deterministic decision tree (the worst-case number of classical queries of an exact classical algorithm). In the worst-case convention used here $D = R_0$: fixing the random string of a zero-error randomized protocol that never exceeds T queries leaves a deterministic protocol that is correct on every instance with at most T queries. (If zero-error cost is instead measured in expectation, as is common, it can be smaller than D.) Classically, a deterministic protocol partitions Ω by its possible transcripts and succeeds exactly only when that transcript partition refines Π_g. A quantum protocol need not create such an intermediate classical transcript; it changes the overlaps among the hypothesis states and measures only after the required cells have separated. A speedup is a gap between the quantum and the classical cost for the same task and the same interface.

All Hilbert spaces in this paper are finite dimensional. For a family of unitary oracles $\{U_a : a \in \Omega\}$, an arbitrary one-query protocol prepares a pure state $|\Psi\rangle$ of the query system Q and a reference R, applies $U_a \otimes I_R$, and then uses an instance-independent channel and measurement. Write

$$|\Psi_a\rangle = (U_a \otimes I_R)\,|\Psi\rangle\,, \qquad \sigma = \mathrm{Tr}_R\,|\Psi\rangle\langle\Psi|\,. \tag{3}$$

The output Gram matrix before the final processing is

$$G_{aa'} = \langle\Psi_a|\Psi_{a'}\rangle = \mathrm{Tr}\big(\sigma U_a^\dagger U_{a'}\big)\,. \tag{4}$$

Proposition 1 (Block discrimination)**.** *The partition Π_g is exactly resolvable with at most one use of $\{U_a\}$, allowing an arbitrary reference, if and only if there is a density operator σ such that*

$$\mathrm{Tr}\big(\sigma U_a^\dagger U_{a'}\big) = 0 \quad \textit{whenever} \quad g(a) \neq g(a'). \tag{5}$$

Proof. Exact discrimination requires the spans of the output states associated with different cells to be orthogonal, which is equivalent to the vanishing of the indicated pairwise inner products; no instance-independent channel can make nonorthogonal states perfectly distinguishable. Conversely, orthogonal cell spans can be measured by their support projectors. Purification shows that optimizing over σ already includes mixed probes and arbitrary ancillary systems. □

Equation (5), with $\sigma \succeq 0$ and $\mathrm{Tr}\,\sigma = 1$, is the block form of the standard conditions for perfect discrimination of unitaries [16, 17]; it is not a new general channel-discrimination theorem. Its role here is to connect an answer partition to a specified oracle family. The Gram matrix need only be block diagonal according to Π_g; states within one cell may overlap or coincide. Complete identification is the singleton-cell special case, where G must be diagonal. For every nontrivial partition, zero queries cannot resolve the task, so Eq. (5) characterizes $Q_E(\Pi_g;\mathcal{U}) = 1$. For bounded error, orthogonality is replaced by a common POVM with the required success probability. Multiquery lower bounds come from adversary methods [24], whose generalized form characterizes state conversion and function evaluation [25, 26].

When Eq. (5) fails, a finite certificate says so.

Proposition 2 (Dual certificate)**.** *Let $X = \{(a,a') : g(a) \neq g(a')\}$. Exactly one of the following holds: (i) a density operator σ satisfies Eq. (5); (ii) there is a complex matrix Γ, supported on X and satisfying $\Gamma_{a'a} = \Gamma^*_{aa'}$, such that*

$$Z(\Gamma) = \sum_{(a,a')\in X} \Gamma_{aa'}\, U_a^\dagger U_{a'} \succ 0. \tag{6}$$

TABLE I. Relation to prior work. "Known" lists results available before this paper; "This work" lists what is added here.

Topic	Known	This work
Exact one-query criterion	Perfect discrimination of unitaries [16, 17]	Block form for answer partitions and its dual certificate (Props. 1 and 2); standard convex duality, stated for use below
One-query exact class	XOR oracle: dictators and two-bit parities [14]; partial functions [18]	Every response unitary V: dictators iff $0 \in \mathrm{conv}(\mathrm{spec}\, V)$, two-bit parities iff $-1 \in \mathrm{spec}\, V$, no other tasks (Thm. 1, Prop. 4)
Dependence on the oracle model	Oracle representation [5]; in-place versus standard oracles [4]; de-quantised Deutsch oracle [6]; faulty oracles [7]	Classically invisible spectrum decides exact cost: two completions of one blank-register table, phase-dressed shift (Cor. 1); elementary complement obstruction for sign oracles (Prop. 3)
Bounded error, odd cyclic response	—	Optimal unambiguous one-query solution, worst-case success $(3+\cos\frac{\pi}{d})/4$ (Prop. 5)
Parity	$Q_E(\mathrm{Par}_N) = \lceil N/2 \rceil$ [19, 20]	Degree bound for every response; open qutrit question (Sec. III)
Three-bit functions	Explicit optimal exact algorithms for all functions [15]	No new values; reorganization as balanced partitions (Table II)
Four-bit functions	Numerical SDP values for all functions [15], used in structural work [21]	Exact rational dual certificates for all 6435 balanced partitions (Table III)
Decohered access	Which-way information [22]; coherence in Bernstein–Vazirani [23]	Exact costs with an inaccessible witness: surviving partition in one query versus $n-k$; identification in exactly $\min\{k+1, n\}$ queries (Prop. 6)

Proof. $Z(\Gamma)$ is Hermitian because $(U_a^\dagger U_{a'})^\dagger = U_{a'}^\dagger U_a$. If (i) and (ii) both held, then $\mathrm{Tr}[\sigma Z(\Gamma)] = 0$, contradicting $Z(\Gamma) \succ 0$. If (ii) fails, the real subspace $\{Z(\Gamma)\}$ of Hermitian operators misses the open convex cone of positive definite operators, so a separating hyperplane gives a nonzero σ with $\mathrm{Tr}(\sigma P) \geq 0$ for all $P \succ 0$, that is $\sigma \succeq 0$, and $\mathrm{Tr}[\sigma Z(\Gamma)] = 0$ for all admissible Γ. Taking Γ supported on a single pair (a, a') and its transpose, with real and with imaginary entries, gives $\mathrm{Tr}(\sigma U_a^\dagger U_{a'}) = 0$; normalizing σ gives (i). □

The matrix Γ weights only cross-cell pairs, as an adversary matrix does. After t queries the same weights define the progress measure $\sum_{(a,a')\in X} \Gamma_{aa'} \langle \psi_a^{(t)} | \psi_{a'}^{(t)} \rangle$ of the adversary method [24], and the query cost measures how fast such weighted cross-cell overlaps can be driven to zero. This is the precise sense in which a quantum query algorithm "separates" answer cells. When all $U_a^\dagger U_{a'}$ commute—as for every point oracle below, which is diagonal in the eigenbasis of its response—only the diagonal of σ in a common eigenbasis enters Eq. (5), and the criterion becomes a linear feasibility problem for a probability vector.

The criterion also distinguishes a change of representation from a change of access. If

$$V_a = A U_a B \tag{7}$$

for instance-independent unitaries A, B, then $V_a^\dagger V_{a'} = B^\dagger U_a^\dagger U_{a'} B$; as the probe varies, the attainable Gram matrices are unchanged. More generally, families that exactly simulate one another with $O(1)$ oracle calls have quantum query complexities equal up to constant factors; an approximate simulation must also charge its conversion error to ϵ. Preservation of a claimed quantum–classical gap additionally requires corresponding constant-query classical simulations. Exact one-query behavior can nevertheless change when the response representation is not of the form of Eq. (7), as the next section shows.

III. THE RESPONSE SPECTRUM DECIDES EXACT ONE-QUERY POWER

A. Deutsch's problem and the sign oracle

Let $b = (b_0, b_1)$ be the truth table of a Boolean function and use

$$O_b |x, y\rangle = |x, y \oplus b_x\rangle . \tag{8}$$

With the answer qubit in $|-\rangle$, phase kickback gives, up to the common answer factor, the output states on the address register

$$|\psi_{00}\rangle = -|\psi_{11}\rangle = |+\rangle , \quad |\psi_{01}\rangle = -|\psi_{10}\rangle = |-\rangle . \tag{9}$$

The constant and balanced cells occupy orthogonal rays, while the two instances inside either cell are not distinguished. This is a block-diagonal, but not diagonal,

Gram matrix and gives the familiar one-versus-two exact-query separation [1–3].

Hadamard conjugation of the answer qubit turns Eq. (8) into the controlled-phase action $|x,z\rangle \mapsto (-1)^{zb_x}|x,z\rangle$, so the two forms are exactly equivalent in the sense of Eq. (7). The restricted sign oracle $P_b|x\rangle = (-1)^{b_x}|x\rangle$, without the control qubit z, is weaker: $P_{\bar{b}} = -P_b$ for the complemented truth table $\bar{b}$, so complementary truth tables differ only by a global phase.

Proposition 3 (Complement obstruction)**.** *With access to $\{P_b\}$ alone (arbitrary ancillas, but no control on P_b), a partition in which some b and its complement $\bar{b}$ lie in different cells cannot be resolved with any number of queries, even with error $\epsilon < 1/2$.*

Proof. Each use of $P_{\bar{b}} \otimes I$ equals $-(P_b \otimes I)$, and every branch of an adaptive protocol picks up the same sign. The final density operators for b and $\bar{b}$ are therefore identical, and so are the output distributions. □

Deutsch's constant–balanced partition is closed under complement and remains one-query resolvable with P_b. Three-bit majority, parity of an odd number of bits, and every other self-dual function are not resolvable at all. Among the balanced partitions studied in Sec. IV, only 3 of the 35 three-bit partitions (the two-bit parities) and 35 of the 6435 four-bit partitions are closed under complement; the sign oracle cannot resolve any of the others, whereas the XOR oracle resolves all of them with at most two ($N = 3$) or three ($N = 4$) queries. Supplying a coherent control for P_b restores the missing relative phase, but coherent control is additional black-box access rather than an automatic operation on an unknown unitary [27].

B. Response unitaries

Let V be any unitary and consider the controlled-response family indexed by the same four Boolean truth tables,

$$U_b^{(V)} = \sum_{x=0}^{1} |x\rangle\langle x| \otimes V^{b_x}. \tag{10}$$

Access conventions. The family $\{U_b^{(V)}\}$ is specified, and V is known. Between queries the algorithm may apply any instance-independent operation, including the preparation of eigenvectors of V, at no query cost; it may not control, invert, or otherwise modify the oracle unless such access is specified separately. The phase of V is part of the specification. Because V^{b_x} acts only on the branch selected by the address, replacing V by $e^{i\theta}V$ multiplies Eq. (10) by the instance-dependent phase $e^{i\theta b_x}$, which is relative between address sectors and therefore observable; it can change the spectral condition below (Corollary 1). This differs from access to an unspecified unitary or channel known only up to a global phase.

Theorem 1 (Response spectrum)**.** *The Deutsch constant–balanced partition is exactly one-query resolvable through the family $\{U_b^{(V)}\}$, with arbitrary probes and references, if and only if -1 is an eigenvalue of V.*

Proof. If $V|w\rangle = -|w\rangle$, the probe $|+\rangle|w\rangle$ gives ordinary phase kickback and Eq. (9). Conversely, give weight $\Gamma = 1$ to the four cross-cell pairs (and their transposes) in Proposition 2. The pairs $(00,01)$, $(00,10)$, $(11,01)$, and $(11,10)$ contribute $I \oplus V$, $V \oplus I$, $V^\dagger \oplus I$, and $I \oplus V^\dagger$ on the address blocks $x = 0, 1$, so

$$Z = 2\,I_2 \otimes \left(2I + V + V^\dagger\right) = 2\,I_2 \otimes (I+V)(I+V)^\dagger. \tag{11}$$

Z is positive definite unless $I + V$ is singular, that is, unless -1 is an eigenvalue of V. □

Geometrically, the probe weights on the two address blocks define points of the numerical range $\mathrm{conv}(\mathrm{spec}\,V)$, and cross-cell orthogonality forces both points to -1. Because -1 is an extreme point of the unit disk, it lies in the numerical range only if it is an eigenvalue. Related qudit versions of Deutsch–Jozsa use higher-dimensional address and response systems [2, 3, 28]. Theorem 1 instead holds the four Deutsch instances fixed and varies the response unitary. It is also distinct from prior-distribution "useless-query" bounds, which infer quantum lower bounds from classical queries that reveal no information about a property [29].

Corollary 1 (Reversible completions)**.** *On a qutrit, let T be the transposition $|0\rangle \leftrightarrow |1\rangle$ that fixes $|2\rangle$, let $X_3|y\rangle = |y+1 \bmod 3\rangle$ be the cyclic shift, and let $X_3' = e^{i\pi/3}X_3$. All three map the blank register $|0\rangle$ to $|1\rangle$ up to a phase ($X_3'|0\rangle = e^{i\pi/3}|1\rangle$), so a measurement of a blank target returns the point value b_x in each case; X_3 and X_3' moreover agree on every computational-basis input up to a phase. Their spectra are $\{1,1,-1\}$, $\{1, e^{2\pi i/3}, e^{4\pi i/3}\}$, and $\{e^{i\pi/3}, -1, e^{-i\pi/3}\}$, so the exact quantum cost of Deutsch's problem is $Q_E = 1$, 2, and 1, respectively, while $D = 2$ in all three cases. More generally, for $X_d|y\rangle = |y+1 \bmod d\rangle$ with $d \geq 2$,*

$$Q_E^{(d)} = \begin{cases} 1, & d \text{ even}, \\ 2, & d \text{ odd}, \end{cases} \qquad D^{(d)} = 2. \tag{12}$$

An oracle specified classically by its reversible action on a blank target, $|x\rangle|0\rangle \mapsto |x\rangle|b_x\rangle$, fixes only one column of each V^{b_x}. The completion—transposition or cyclic shift—decides whether Deutsch's exact advantage exists, and so does a phase that no computational basis query can see: the factor $e^{i\pi b_x/3}$ in $X_3'^{b_x}$ is a branch-relative phase, that is, a phase oracle composed with the shift. Thus classical point values, and even the full action on computational basis states up to phases, do not determine Q_E. This is a constant-factor effect, not an asymptotic separation.

C. All point oracles with a response unitary

The same argument applies to any number of bits. Let $f : \{0,1\}^N \to \{0,1\}$ be nonconstant and accessed through $U_b^{(V)} = \sum_{i=1}^N |i\rangle\langle i| \otimes V^{b_i}$.

Proposition 4 (One-query class)**.** *A nonconstant f is exactly one-query resolvable through $\{U_b^{(V)}\}$ if and only if, up to negation, either $f = b_i$ and $0 \in \mathrm{conv}(\mathrm{spec}\, V)$, or $f = b_i \oplus b_j$ and $-1 \in \mathrm{spec}\, V$. The dictator condition holds in particular whenever $V|y\rangle \perp |y\rangle$ for some basis state $|y\rangle$.*

Proof. Let $\rho_i \succeq 0$ be the diagonal block of σ at address i and $p_i = \mathrm{Tr}\,\rho_i$, so $\sum_i p_i = 1$. Since $U_b^\dagger U_{b'} = \sum_i |i\rangle\langle i| \otimes V^{b'_i - b_i}$, only the ρ_i enter Eq. (5). If f depends on b_i, some b has $f(b) \neq f(b \oplus e_i)$, and Eq. (5) reads $1 - p_i + \mathrm{Tr}(\rho_i V^{\pm 1}) = 0$. Since $|\,\mathrm{Tr}(\rho_i V^{\pm 1})| \le p_i$, this forces $p_i \ge 1/2$, so f depends on at most two bits. If it depends on two, i and j, then $p_i = p_j = 1/2$ and $\mathrm{Tr}(\rho_i V^{\pm 1}) = -p_i$, which places ρ_i and ρ_j in the -1 eigenspace of V. The overlap of the outputs for b and $b \oplus e_i \oplus e_j$ is then $-1 \neq 0$, so these instances share a cell, $f(b) = f(b \oplus e_i \oplus e_j)$, and f is $b_i \oplus b_j$ or its negation. A -1 eigenvector gives the Deutsch construction on addresses i, j. For a dictator b_i, the cross-cell pairs $(0^N, e_i)$ and $(0^N, e_i \oplus e_j)$, $j \neq i$, give $\mathrm{Tr}(\rho_i V) + \sum_{j \neq i} p_j = 0$ and $\mathrm{Tr}(\rho_j V) = p_j$, so the density operator $\tau = \sum_j \rho_j$ has $\mathrm{Tr}(\tau V) = 0$, that is, $0 \in \mathrm{conv}(\mathrm{spec}\, V)$. Conversely, the probe $|i\rangle\,|w\rangle$ with $\langle w|V|w\rangle = 0$ makes every cross-cell overlap vanish. With an additional null address, on which the oracle acts as the identity (as in the phase form used in Sec. IV), the same argument holds with $\mathrm{spec}\, V$ replaced by $\mathrm{spec}\, V \cup \{1\}$ in the dictator condition. □

For $V = -I$, the family is the sign oracle of Proposition 3: two-bit parities are one-query resolvable, whereas dictators are not resolvable at all.

For the XOR oracle, $V = X$, Proposition 4 reproduces the characterization of exact one-query algorithms by Chen, Ye, and Li [14]; partial functions are treated in Ref. [18]. For an odd cyclic response X_d the one-query class shrinks to the dictators, which a single basis query already resolves classically.

D. Bounded error for odd cyclic responses

Corollary 1 concerns exact resolution. With bounded error the odd cyclic response keeps a quantum advantage. As a referee pointed out, a single eigenvector of X_3 with eigenvalue $e^{2\pi i/3}$, the address in $|+\rangle$, and a measurement of the address in $\{|+\rangle, |-\rangle\}$ already give success 1 on constant and 3/4 on balanced instances, whereas one classical point query cannot exceed worst-case success 1/2. Spreading the probe over the two eigenvalues adjacent to -1 raises the worst case further and makes every conclusive answer correct.

Proposition 5 (Optimal one-query solution)**.** *For odd $d \ge 3$, one query to $\{U_b^{(X_d)}\}$ solves Deutsch's problem with zero error and a conclusive answer with probability $(1 + \cos\frac{\pi}{d})/2$ on every instance. Guessing on the inconclusive outcome gives worst-case success $(3 + \cos\frac{\pi}{d})/4$, that is 3/4 conclusive and 7/8 overall for $d = 3$, and no one-query protocol does better. One classical query never gives a conclusive zero-error answer and has worst-case success 1/2.*

Proof. Let $|v_\pm\rangle$ be eigenvectors of X_d with the eigenvalues $\lambda_\pm = -e^{\pm i\pi/d}$ adjacent to -1, and use the probe $\frac{1}{2}\sum_{x=0}^{1}\sum_\pm |x\rangle\,|v_\pm\rangle$. Writing $\gamma = \cos\frac{\pi}{d}$ and using $(\lambda_+^k + \lambda_-^k)/2 = -\gamma$ for $k = \pm 1$, the output Gram matrix is $G_{bb'} = \frac{1}{2}\sum_x c_x$ with $c_x = 1$ if $b_x = b'_x$ and $c_x = -\gamma$ otherwise. Hence $|\psi_{00}\rangle + |\psi_{11}\rangle = |\psi_{01}\rangle + |\psi_{10}\rangle$, with squared norm $2(1-\gamma)$, while $|\psi_{00}\rangle - |\psi_{11}\rangle$ and $|\psi_{01}\rangle - |\psi_{10}\rangle$ are orthogonal to each other and to that common vector, each with squared norm $2(1+\gamma)$. Projecting onto these three directions returns "constant," "balanced," or "inconclusive"; each instance gives the correct conclusive outcome with probability $(1+\gamma)/2$ and never the wrong one.

For optimality, write any pure probe, including a reference, as $|0\rangle\,|A\rangle + |1\rangle\,|B\rangle$ with $\|A\|^2 + \|B\|^2 = 1$, and put $W = V \otimes I_R$ with $V = X_d$. Let ρ_C and ρ_B be the outputs averaged uniformly over the constant and over the balanced instances. Their diagonal address blocks coincide, and their off-diagonal blocks differ by $\frac{1}{2}\,|(I-W)A\rangle\langle(I-W)B|$, so $\|\rho_C - \rho_B\|_1 = \|(I-W)A\|\,\|(I-W)B\| \le \frac{1}{2}\|I - V\|^2$. Worst-case success cannot exceed success under this prior, which by the Helstrom bound is at most $\frac{1}{2} + \frac{1}{8}\|I - V\|^2$. For odd d, $\|I - X_d\|^2 = 2 + 2\cos\frac{\pi}{d}$, which gives $(3 + \cos\frac{\pi}{d})/4$. Classically, one query reveals one bit b_x, and both cells remain consistent with every transcript. □

The bound $\frac{1}{2} + \frac{1}{8}\|I - V\|^2$ holds for every response V and reaches 1 only if $\|I - V\| = 2$, that is, only if -1 is an eigenvalue; this is a second proof of the necessity part of Theorem 1. The construction parallels Deutsch's original algorithm, which also answered conclusively only part of the time [1]; here the inconclusive probability $(1 - \cos\frac{\pi}{d})/2$ is set by the distance of the spectrum from -1 and vanishes as $d \to \infty$.

E. Parity

Proposition 4 shows that one query resolves the parity of N bits only for $N \le 2$. Proposition 2 gives an explicit witness for the XOR oracle: with $\Gamma = 1$ on the pairs $(0^N, e_i)$,

$$Z = \sum_{i=1}^{N} \left(O_{e_i} + O_{e_i}^\dagger\right) = 2N\, I - 4\Pi_-, \tag{13}$$

where $\Pi_- = \sum_i |i\rangle\langle i| \otimes |-\rangle\langle -|$; Z is positive definite exactly when $N > 2$. For more queries, note that $U_b^{(V)} =$

$\sum_i |i\rangle\langle i| \otimes [I + b_i(V - I)]$ is affine in b for every response V. The amplitudes after T queries are therefore polynomials of degree at most T in the input bits, the acceptance probability has degree at most $2T$, and parity has exact degree N. Thus $T \geq N/2$ for every response, and for the XOR oracle applying the Deutsch construction to disjoint pairs gives the matching algorithm, so

$$Q_E(\mathrm{Par}_N) = \lceil N/2 \rceil, \qquad D(\mathrm{Par}_N) = N \tag{14}$$

for point-value access [19, 20]. For the qutrit shift X_3, Theorem 1 forbids the pair construction, and $\lceil N/2 \rceil \leq Q_E(\mathrm{Par}_N; X_3) \leq N$. Whether the upper bound is attained is open; if it is, two oracles with identical classical point values would differ by a factor of two for all N.

If instead one declares $\widetilde{O}_b |y\rangle = |y \oplus \mathrm{Par}(b)\rangle$ to be a single oracle call, parity becomes one-query resolvable classically as well as quantum mechanically. That does not contradict Eq. (14); it replaces point access by an oracle that already aggregates the desired answer.

IV. EXACT CENSUS OF BALANCED POINT-ORACLE TASKS

The preceding examples vary the access while holding the task fixed. We now hold the standard point oracle fixed and vary the task. Let the hidden instance be $b \in \{0,1\}^N$, accessed through Eq. (8) with the address range enlarged to N, and consider every binary answer map with 2^{N-1} instances in each cell. Exchanging the answer labels gives the same unlabeled partition, so their number is

$$M_N = \frac{1}{2}\binom{2^N}{2^{N-1}}. \tag{15}$$

Coordinate permutations, independent complements of coordinates, and answer exchange preserve both D and Q_E. They reduce the $M_3 = 35$ partitions to 6 orbits and the $M_4 = 6435$ partitions to 58. Orbit sizes are restored when reporting the counts below.

Methods. The decision-tree depth D was computed exactly by dynamic programming over subcubes. By Proposition 4 (Ref. [14]), $Q_E = 1$ holds exactly for dictators and two-bit parities. Upper bounds on Q_E come from adaptive trees whose nodes read either one bit or the parity of two bits, each costing one query; such trees are exact upper certificates but not a general characterization of exact quantum algorithms [15]. Lower bounds $Q_E \geq 3$ come from the two-query semidefinite program of Barnum, Saks, and Szegedy [30]. We use its bounded-error form, which is better conditioned than an exact feasibility test. In the phase form of the point oracle, $|i\rangle \mapsto (-1)^{b_i} |i\rangle$ with a null address $i = 0$ and $b_0 = 0$, put $D_i = \mathrm{diag}_b\big((-1)^{b_i}\big)$ (so $D_0 = I$). A two-query algorithm with worst-case success p yields positive semidefinite $2^N \times 2^N$ matrices with $\sum_i A_i = J$ (the all-ones matrix), $\sum_i D_i A_i D_i = \sum_i B_i$, $\sum_i D_i B_i D_i = \Gamma_0 + \Gamma_1$, and $(\Gamma_{f(b)})_{bb} \geq p$. If real symmetric Y_1, Y_2, Y_3 and a probability vector λ satisfy

$$\begin{aligned} &Y_1 \succeq D_i Y_2 D_i, \quad Y_2 \succeq D_i Y_3 D_i \quad (i = 0, \ldots, N), \\ &Y_3 \succeq \Lambda_j \quad (j = 0, 1), \end{aligned} \tag{16}$$

with $\Lambda_j = \mathrm{diag}_b\big(\lambda_b [f(b) = j]\big)$, then

$$\begin{aligned} \mathbf{1}^{\mathsf{T}} Y_1 \mathbf{1} &= \textstyle\sum_i \langle Y_1, A_i \rangle \geq \langle Y_2, \sum_i B_i \rangle \geq \sum_j \langle Y_3, \Gamma_j \rangle \\ &\geq \textstyle\sum_b \lambda_b (\Gamma_{f(b)})_{bb} \geq p, \end{aligned} \tag{17}$$

so $\mathbf{1}^{\mathsf{T}} Y_1 \mathbf{1} < 1$ proves $Q_E \geq 3$. We solved Eq. (16) numerically with a barrier method, added 3ε, 2ε, and ε times the identity to Y_1, Y_2, and Y_3 with $\varepsilon = 2^{-20}$, which raises every constraint matrix in Eq. (16) by εI, rounded every entry to a dyadic rational with denominator 2^{40}, and verified all twelve matrix inequalities and $\lambda \geq 0$, $\sum_b \lambda_b = 1$ in exact integer arithmetic (positive pivots in Gaussian elimination). Each certificate is a list of integers, given in the Supplemental Material, and can be checked independently of the solver.

Use of AI tools. The census code and the Lean formalization were written with an OpenAI Codex assistant, and the dual-certificate solver and the exact verifier with Anthropic's Claude (versions are given in the Acknowledgments). No reported result rests on these tools alone: the census counts were reproduced by two independently written programs and agree with Ref. [15], the Lean audits reproduce the finite census, and every certificate is checked by a verifier that uses only exact integer arithmetic and does not depend on the solver.

Three bits. For $N = 3$ the one-query class and explicit trees meet for every task (Table II). These exact values are known: Montanaro, Jozsa, and Mitchison give optimal exact algorithms for every three-bit function [15], and the contribution here is only their organization as balanced partitions. Thus 20 of the 35 balanced tasks admit an exact advantage: three by a factor of two and seventeen by a factor of 3/2. The trees display the conditional structure hidden by the bare partition. Three-bit majority is decided by first obtaining $b_1 \oplus b_2$; if it is zero one reads b_1, and if it is one one reads b_3; this is the

TABLE II. Balanced three-bit tasks up to coordinate permutations, input complements, and answer exchange. The orbit sizes sum to 35. Here $\mathrm{Sel}(c, u, v) = u$ for $c = 0$ and v for $c = 1$. The values of Q_E are not new: Ref. [15] gives explicit optimal exact algorithms for every function on up to three bits.

Representative	Orbit	D	Q_E
b_1	3	1	1
$b_1 \oplus b_2$	3	2	1
Par_3	1	3	2
Maj_3	4	3	2
$b_1 \oplus b_2 b_3$	12	3	2
$\mathrm{Sel}(b_1, b_2, b_3)$	12	2	2

TABLE III. Balanced four-bit tasks grouped by (D, Q_E). Orbits are taken up to coordinate permutations, input complements, and answer exchange; the columns sum to 58 orbits and 6435 partitions. The last column names the source of the lower bound on Q_E: Proposition 4 excludes one query, and the rational dual certificates of Eq. (16) exclude two.

(D, Q_E)	Orbits	Partitions	Representative	Lower bound
$(1,1)$	1	4	b_1	—
$(2,1)$	1	6	$b_1 \oplus b_2$	—
$(2,2)$	1	48	$\mathrm{Sel}(b_1, b_2, b_3)$	Prop. 4
$(3,2)$	8	248	$\mathrm{Maj}(b_1, b_2, b_3)$	Prop. 4
$(4,2)$	4	89	Par_4	Prop. 4
$(3,3)$	10	1536	$\mathrm{Sel}(b_1, b_2 b_3, b_2 \vee b_4)$	certificate
$(4,3)$	33	4504	$b_1 \oplus b_2 \oplus b_3 b_4$	certificate

algorithm of Ref. [15], Sec. 6.2. Likewise, $F = b_1 \oplus b_2 b_3$ is decided by first reading b_2, followed by b_1 when $b_2 = 0$ and by the Deutsch parity $b_1 \oplus b_3$ when $b_2 = 1$.

Four bits. For $N = 4$ the one-query class and explicit two-query trees settle 15 orbits covering 395 partitions. For each of the remaining 43 orbit representatives, covering 6040 partitions, explicit trees give $Q_E \le 3$ and the exact rational certificates bound the optimal two-query worst-case success by values between 0.9001 and 0.9862, hence $Q_E = 3$. The classification of Table III is therefore exact: 4847 balanced four-bit partitions have an exact quantum advantage (6 by a factor of two at $(2,1)$, 89 by a factor of two at $(4,2)$, 248 by a factor of $3/2$, and 4504 by a factor of $4/3$) and 1588 have none. The polynomial method could not have supplied the lower bounds for the 43 orbits: their real degrees are 3 (2144 partitions) or 4 (3896 partitions), so it yields only $Q_E \ge 2$.

Numerical SDP values for all Boolean functions through four bits were reported in Ref. [15], whose authors note that such values do not by themselves certify exactness; subsequent structural work on exact two-query algorithms uses that four-bit classification [21]. We cross-validated our orbit representatives against the published NPN (negation–permutation–negation) identifiers of Ref. [15]. After removing dummy variables, its one- and two-query entries agree with all 15 representatives settled by trees, and for the 43 remaining representatives its optimal two-query success probabilities, between 0.900 and 0.986, agree with the optimal values 0.9000–0.9861 of our dual solutions. The certificates turn that numerical agreement into exact lower bounds, and the 15 orbits settled by trees give dual optimal value one, as they must.

Machine-checked companion. The Supplemental Material includes a Lean 4/mathlib formalization of the exact algebraic and finite core. Three `native_decide` audits reproduce the 35-task and 6435-task deterministic and restricted-tree censuses, including the 395 cases settled by trees; `native_decide` additionally trusts Lean's compiler and native evaluator through an extra axiom, which `#print axioms` reports. The 43 dual certificates are checked by a separate exact-arithmetic verifier, not yet in Lean. Density-operator semantics, unrestricted query semantics, and the general parity lower bound remain unformalized.

The contribution of the census is to organize balanced functions as unlabeled answer partitions, quotient only by symmetries that preserve point access, and attach a short exact certificate to every class. The observed proportions are finite-model facts, not an asymptotic speedup criterion.

V. DECOHERED ACCESS: BERNSTEIN–VAZIRANI WITH AN INACCESSIBLE WITNESS

For $a, x \in \mathbb{F}_2^n$, let $f_a(x) = a \cdot x \pmod 2$ and $O_a |x, y\rangle = |x, y \oplus a \cdot x\rangle$. With ordinary phase access, the corresponding character states are orthogonal, so the singleton partition is resolved in one query [31]. We now add an inaccessible address record while preserving every computational-basis value returned by the oracle.

Choose $S \subseteq \{1, \dots, n\}$, $|S| = k$, let x_S denote the corresponding substring, and let $\bar{S}$ be the complement of S. Consider the unitary dilation

$$W_a^S |x, y, e\rangle = |x, y \oplus a \cdot x, e \oplus x_S\rangle . \tag{18}$$

Each call receives a fresh witness in $|0^k\rangle$, which is discarded afterward. The accessible oracle is consequently the channel

$$\Phi_a^S(\rho) = \mathrm{Tr}_E\left[W_a^S(\rho \otimes |0^k\rangle\langle 0^k|)(W_a^S)^\dagger\right] = O_a\, \Delta_S(\rho)\, O_a^\dagger, \tag{19}$$

where $\Delta_S(\rho) = \sum_s \Pi_s \rho \Pi_s$ dephases the address register in the coordinates S (Π_s projects onto $x_S = s$); the second form holds because copying x_S into a discarded register is dephasing, and O_a commutes with every Π_s. Basis queries still return $a \cdot x$ exactly, while the witness removes coherences between sectors with different x_S, as in standard which-way interference [22] and in the coherence analysis of the Bernstein–Vazirani algorithm [23].

Surviving partition. Define

$$\Pi_S = \{C_u : u \in \mathbb{F}_2^{n-k}\}, \qquad C_u = \{a : a_{\bar{S}} = u\}. \tag{20}$$

For the classical comparison, let $\mathcal{O}^{\mathrm{val}} = \{O_a^{\mathrm{val}} : a \in \mathbb{F}_2^n\}$ denote the value-oracle family induced by computational-basis preparation and readout of Φ_a^S: a query chooses x and returns $a \cdot x$. For $k < n$ and worst-case error $\epsilon < 1/2$, these interfaces give

$$Q_\epsilon^\Phi(\Pi_S) = 1, \qquad R_\epsilon(\Pi_S; \mathcal{O}^{\mathrm{val}}) = n - k. \tag{21}$$

For the quantum part, the usual Bernstein–Vazirani input and phase kickback, followed by Eq. (19), leave on the n-qubit address register (after factoring out the answer qubit, which stays in $|-\rangle$) the state

$$\rho_a^S = 2^{-n} \sum_{x,x' :\, x_S = x'_S} (-1)^{a\cdot(x\oplus x')} \, |x\rangle\langle x'| \, . \tag{22}$$

Character orthogonality after the final Hadamards gives, up to a fixed reordering of coordinates,

$$\begin{aligned} H^{\otimes n} \rho_a^S H^{\otimes n} &= |a_{\bar S}\rangle\langle a_{\bar S}| \otimes \frac{I_S}{2^k}, \\ \Pr(z|a) &= 2^{-k}\, \mathbf{1}[z_{\bar S} = a_{\bar S}]. \end{aligned} \tag{23}$$

Thus one query exactly resolves Π_S; zero queries cannot, because the output would be independent of the cell label. The classical upper bound uses the basis queries outside S. For the lower bound, impose the favorable promise $a_S = 0$. Under the uniform distribution on $a_{\bar S}$, fewer than $n-k$ returned linear equations leave an affine solution space of positive dimension, so even a decoder given the complete transcript succeeds with probability at most $1/2$. The same average bound holds after including the protocol's internal randomness, and therefore some input has success at most $1/2$.

Original task. The same channel also changes the cost of the original singleton task, identification of a.

Proposition 6 (Witness cost of identification)**.** *For $0 \le k \le n$, with S known to the algorithm, the exact quantum cost of identifying a through $\{\Phi_a^S\}$ is $Q_E^\Phi(\Pi_{\mathrm{single}}) = \min\{k+1, n\}$; classically, $R_\epsilon(\Pi_{\mathrm{single}}; \mathcal{O}^{\mathrm{val}}) = n$ for $\epsilon < 1/2$.*

Proof. Upper bound: one Bernstein–Vazirani query with $x_S = 0$ returns $a_{\bar S}$, and k basis queries $x = e_i$, $i \in S$, return a_S; alternatively, n basis queries return a. At $k = 0$ this is the ordinary one-query algorithm. Lower bound for $k \ge 1$: record each dephasing in Eq. (19) in an inaccessible environment register. For a record $\vec{s} = (s_1, \dots, s_T)$, $s_t \in \mathbb{F}_2^k$, let $|\phi_a(\vec{s})\rangle$ be the conditional unnormalized output, so that $\rho_a = \sum_{\vec{s}} |\phi_a(\vec{s})\rangle\langle\phi_a(\vec{s})|$. Because the records are inaccessible, exact identification requires $\langle\phi_a(\vec{s})|\phi_{a'}(\vec{s}')\rangle = 0$ for all $a \ne a'$ and all records $\vec{s}, \vec{s}'$. In the answer qubit's X basis, the tth query multiplies the component at address (s_t, z) either by 1 or by $(-1)^{a\cdot(s_t,z)}$. Hence $\phi_a(\vec{s})$ is a linear combination of characters $(-1)^{a\cdot f}$ whose frequencies are $f = 0$ or $f = \sum_{t\in F}(s_t, z_t)$ for nonempty subsets F of the queries. (i) If $s_1, \dots, s_T$ do not span $\mathbb{F}_2^k$, choose $h \ne 0$ orthogonal to all s_t. Every such character is then invariant under $a \mapsto a + (h, 0)$, so $\phi_{a+(h,0)}(\vec{s}) = \phi_a(\vec{s})$, and orthogonality forces $\phi_a(\vec{s}) = 0$. Hence $T \ge k$, and only records whose s_t span $\mathbb{F}_2^k$ contribute. (ii) Let $T = k < n$. In every contributing record the s_t are linearly independent, so every frequency with $F \ne \emptyset$ has a nonzero S part, and for $u \ne 0$ the sum $\sum_a (-1)^{u\cdot a_{\bar S}} \phi_a(\vec{s})$ vanishes. Since the $\phi_a(\vec{s})$ are pairwise orthogonal, the squared norm of this sum is $\sum_a \|\phi_a(\vec{s})\|^2$, so $\phi_a(\vec{s}) = 0$ for every contributing record, contradicting normalization. Hence $T \ge k+1$ for $k < n$. The classical bound is the singleton case of the argument above. □

At $k = 0$, no address coordinate is recorded and Eq. (23) is the ordinary Bernstein–Vazirani output, giving the n-versus-one identification gap. Each recorded address bit adds exactly one query until the cost reaches the classical value n; from $k = n-1$ on, the quantum advantage for identification is gone, although ordinary basis queries still recover a. If the witness were coherently available, the fixed coupling $|x, e\rangle \mapsto |x, e \oplus x_S\rangle$ could be undone without another oracle call, restoring the ordinary output. The accessible oracle here is a channel rather than the globally unitary dilation: as with in-place versus standard oracles [4] and faulty oracles [7], whether the oracle leaves an inaccessible record is part of the problem specification.

VI. DISCUSSION

All results of this paper follow one strategy, which can be stated in four steps; the physics enters in the middle two.

- *Answer partition* Π_g [Eq. (1)]: Put two hidden instances in the same cell whenever the task has the same correct answer for both. The cells C_j are sets of classical instances, not subspaces. The algorithm must learn the cell, and only the cell: in Deutsch's problem the four possible functions fall into two cells, constant and balanced, and the algorithm never has to find out which function within a cell it faces.

- *Oracle access* (Secs. III and V): Fix what one oracle call does. The partition fixes the correct answers to the question; the access model fixes how that answer can be obtained by quantum state discrimination.

- *Probe and one quantum oracle call* (Sec. II): Choose an instance-independent probe state, for instance a uniform superposition obtained by "fanning out" a single basis state with Hadamard gates. One oracle call turns the probe into the output state $|\Psi_a\rangle$ of the hidden instance a.

- *Orthogonality across cells* (Proposition 1): Ask whether some probe makes $\langle\Psi_a|\Psi_{a'}\rangle = 0$ whenever $g(a) \ne g(a')$ [Eq. (5)]. A negative answer needs

no search over the infinitely many probes: a single matrix, a dual certificate (Proposition 2), proves that none works. Only orthogonal states can be told apart with certainty, so exactness requires orthogonality, but only between cells. Within a cell, states may overlap or even coincide: the measurement reads the cell label, not the instance, as a post office sorts letters by postal code without reading the street.

Two equally weighted branches of a quantum state cancel, that is, interfere destructively, only if one carries an extra factor -1 relative to the other. In the examples of the Introduction, the standard flip on a qubit and the transposition on a qutrit both have the eigenvalue -1 and separate constant from balanced functions in one call; the qutrit cyclic shift, whose eigenvalues are the cube roots of unity, cannot (Theorem 1 and Corollary 1). With more calls, the same orthogonality is required of the final states, with instance-independent operations between the calls (Sec. IV); with bounded error, it is relaxed to a required success probability (Proposition 5).

The separations found here need not cross the stabilizer boundary. The standard Boolean Deutsch and linear Bernstein–Vazirani protocols, including their XOR oracles, use only Clifford operations, and the witness coupling in Eq. (18) is Clifford before the witness is discarded. They are therefore efficiently simulable when a full classical circuit description is supplied [32, 33]. In the query model, however, such a description would reveal hidden oracle data and is stronger than matched black-box value access. For these examples, nonstabilizer resources ("magic" [34]) are not needed for a query advantage. General point oracles with a binary-encoded address require multiply controlled gates and are not Clifford, so this remark is specific to Deutsch and Bernstein–Vazirani.

The results vary these steps one at a time. Theorem 1, Corollary 1, and Propositions 3–6 change the access (the second step) while holding the task and its classical point values fixed, and the exact cost changes. The census holds the point oracle fixed and varies the balanced task (the first step), and the cost changes again, with every class certified exactly for $N \leq 4$. Shor's and Grover's algorithms [35, 36] fit the same perspective: each exploits structure in the oracle family that is matched to a specific coherent access.

Partitions are therefore indispensable as specifications but insufficient as complexity criteria, and so are classical point values. A quantum advantage occurs when structure-matched interference separates the required answer cells in fewer calls than any classical transcript under the same interface. Superlinear exact separations for total Boolean functions are known [37], but establishing any asymptotic speedup requires a structured family of task–oracle pairs; percentages from a finite census cannot substitute for that analysis. Within the present framework, the parity question for the qutrit shift in Sec. III is a natural test of whether a response-spectrum effect persists uniformly in N.

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Data and code availability.—See Supplemental Material at [URL will be inserted by publisher] for the Python and Lean 4 sources, the proof report, the 43 rational dual certificates with an exact verifier, the census tables and witness files, and the cross-validation against the published NPN data of Ref. [15].

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